BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 105 · Abstraction and Decomposition

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does lossy compression do?

Q2. What does RAM hold while a program runs?
A. Active instructions and data
B. Only permanent file backups
C. Only startup firmware
D. Only printed documents

Q3. “Sample rate is the number of bits used for each sample.” Correct this. Explain why.

Q4. Two recordings have the same sample rate and length but different bit depths. “Bit depth is samples per second.” Correct this. How does bit depth affect detail and size?

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I Do · Read new knowledge

Key vocabulary

Decomposition — Breaking a problem into smaller parts.

Abstraction — Removing irrelevant details while retaining essential ones.

Algorithmic thinking — Developing a precise step-by-step solution.

Subproblem — Smaller task that contributes to the overall solution.

New learning

Decomposition: Break a complex problem into smaller connected subproblems, each of which can be solved and tested.

Abstraction: Keep details relevant to the task and ignore irrelevant features.

Algorithmic thinking: Describe a clear sequence of unambiguous steps for solving each subproblem.

Build and test: Combine tested parts and check that the complete solution meets the original requirements.

Worked example

A school wants a library loan system. Demonstrate decomposition and abstraction for this task. [6 marks]

1. Describe the main problem

2. Students must search for and borrow books, while staff track returns.

3. Decompose into search catalogue, check availability, record loan and record return.

4. Abstract away cover colour or shelf decorations; retain book ID, borrower ID and due date. [6]

We Do · Practise with guidance

Guided 1

Decompose an online ticket-booking system into three tasks. [3]

Worked method: Display available tickets [1], accept and validate a booking [1], record payment or issue confirmation [1].

Guided 2

Explain two pieces of information to retain in a parcel-tracking system and one to ignore. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Retain parcel ID [1] and delivery status/location [1]; ignore driver's favourite music [1].

Guided 3

Explain why decomposition makes a complex program easier to develop. [4]

Worked method: Smaller modules are easier to understand [1] and assign to developers [1]; individual parts can be tested [1], making errors easier to isolate [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete this reusable function for the number of raw bitmap bits.

```
function imageBits(width,height,depth)
    return ...
endfunction
print(imageBits(10,4,8))
```

write

Question 1

Define abstraction in computational thinking. [2] Source: Original OCR J277-style practice question.

Question 2

A school creates an attendance app. Suggest THREE suitable subproblems. [3] Source: Original OCR J277-style practice question.

Question 3

Explain how decomposition and abstraction differ. [4] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

A trace table can help check how values change during a program.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It discards some information, so the exact original cannot necessarily be restored.

Q2. A — Active instructions and data

Q3. Sample rate is samples per second. At 8,000 Hz, 8,000 samples are taken each second. A higher rate can capture more detail but uses more space.

Q4. Bit depth is bits per sample. More bits allow more amplitude levels but use more space. Longer duration adds samples; sample rate and bit depth control detail.

I Do / We Do pages · Guided practice

Q1. Display available tickets [1], accept and validate a booking [1], record payment or issue confirmation [1].

Q2. Retain parcel ID [1] and delivery status/location [1]; ignore driver's favourite music [1].

Q3. Smaller modules are easier to understand [1] and assign to developers [1]; individual parts can be tested [1], making errors easier to isolate [1].

You Do pages · Code First

```
function imageBits(width,height,depth)
    return width * height * depth
endfunction
print(imageBits(10,4,8))
```

Expected result: 320

You Do pages · Independent questions

Q1. Focusing on relevant detail [1] and ignoring unnecessary information [1].

Q2. Accept/register pupil status [1], store attendance record [1], calculate/generate an attendance report [1]. Accept valid equivalents.

Q3. Decomposition splits a large problem into smaller parts [1] to manage complexity [1]. Abstraction hides/removes irrelevant detail [1] to focus on what

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matters [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: